
Problem Set 4 Solutions

1. **(6 points—3 each)** On the midterm exam in introductory statistics, an instructor always gives a grade of B to students who score between 80 and 90. The scores tend to have a normal distribution with a mean $\mu = 83$ and a standard deviation $\sigma = 5$. About what fraction of the students get a B?
- (a) First, answer this question using what you know about the normal distribution.
- (b) Now use simulated data in Stata. Generate 1,000 student exam scores—this instructor has a big class!—from a normal distribution with the above parameters. Then answer the question based on the data you drew. Are there any differences between your two answers?

This question is asking: $P(80 \leq X \leq 90) = P(\frac{80-83}{5} \leq \frac{X-\mu}{\sigma} \leq \frac{90-83}{5}) = P(-0.6 \leq z \leq 1.4)$. For part (a), we can use Stata:

`display normal(1.4)-normal(-0.6)`, this probability is 0.645. Or, about 64.5% of students get a B. See the attached do-file for part (b). The proportion scoring between 80 and 90 (inclusive) will generally differ from 0.645 because this is a random sample of 1,000.

2. **(8 points—2 each)** Suppose the SAT math scores of high school seniors follow a normal distribution with mean $\mu = 550$ and standard deviation $\sigma = 100$. Now suppose you take a sample of 25 seniors and calculate the sample mean ($\bar{x}$).
- (a) What is $E(\bar{x})$? What is $Var(\bar{x})$? What is the *standard error* of $\bar{x}$?

$$\begin{aligned}E(\bar{x}) &= \mu = 550 \\Var(\bar{x}) &= \sigma^2/n = 100^2/25 = 400 \\se(\bar{x}) &= \sigma/\sqrt{n} = 100/\sqrt{25} = 20\end{aligned}$$

- (b) Suppose you would like to increase precision and cut your standard error in half. What sample size would you need to use?

Currently the standard error of $\bar{x}$ is 20 with a sample size of 25. If we would like the standard error cut in half to 10, the sample size n will need to satisfy: $100/\sqrt{n} = 10$ or $n=100$. In general, with a standard deviation of σ and a desired standard error of se , the required sample size will be: $n = \sigma^2/se^2$.

- (c) Based on the original sample size of 25, what is the probability that you draw a random sample with a $\bar{x}$ of 590 or higher?

Here we can apply our knowledge that SAT scores follow a normal distribution. The Central Limit Theorem says that if SAT scores are normal, then $\bar{x}$ drawn from a random sample of SAT scores will have a normal sampling distribution, with $E(\bar{x}) = \mu$ and $se(\bar{x}) = \sigma/\sqrt{n}$. So:

$$\begin{aligned} P(\bar{x} \geq 590) &= P\left(\frac{\bar{x} - 550}{100/\sqrt{25}} \geq \frac{590 - 550}{100/\sqrt{25}}\right) \\ &= P(z \geq 2) \\ &= 0.023 \end{aligned}$$

Or 2.3%. The last calculation comes from a statistical table or Stata (`display 1-normal(2)`)

- (d) Based on the original sample size of 25, what is the probability what you draw a random sample with a $\bar{x}$ between 525 and 575?

By the same logic as part c:

$$\begin{aligned} P(525 \leq \bar{x} \leq 575) &= P\left(\frac{525 - 550}{100/\sqrt{25}} \leq \frac{\bar{x} - 550}{100/\sqrt{25}} \leq \frac{575 - 550}{100/\sqrt{25}}\right) \\ &= P(-1.25 \leq z \leq 1.25) \\ &= 0.789 \end{aligned}$$

Or 78.9%. The last calculation comes from a statistical table or Stata (`display normal(1.25)-normal(-1.25)`)

3. (4 points) A national survey conducted in June 2021 by the University of Southern California as part of its Understanding Coronavirus in America program asked participants whether they had received at least one dose of the coronavirus vaccine. Of 1,626 adults interviewed, 67.58% said *yes*. Find the estimated standard error for the sample proportion ($\hat{\pi}$) reporting they had received at least one dose of the vaccine. Interpret this in words. Hint: use the sample proportion in place of the *population* proportion (π) where required.

Recall the standard error of a sample proportion is $\sqrt{(\pi(1 - \pi))/n}$. Since π is unknown we can substitute in our best estimate of π , which is $\hat{\pi}$. In this case the standard error is $\sqrt{(0.6758 * 0.3242)/1626} = 0.0116$. In any given random sample of 1,626 persons, the proportion who answer “yes” to this question will vary. The standard error 0.0116 is a measure of how much the proportion answering “yes” varies from sample to sample.

4. (5 points) Mr. Grumpy and Mr. Happy are both running for Governor. Mr. Grumpy will eventually win the election with 51 percent of the vote. A day before the election, a state-wide newspaper surveys 100 likely voters about their choice for governor. Assume survey respondents accurately report who they will vote for. What is the probability Mr. Happy will be supported by 51 percent or more of the survey respondents?

Knowing that Mr. Grumpy will eventually win with 51 percent of the vote means $\pi = 0.51$ —the true population proportion of voters who favor Grumpy is 0.51. A random sample of 100 voters will provide the *sample* proportion $\hat{\pi}$, an estimate of the true population proportion. The standard error of a sample proportion across repeated samples is $\sqrt{(\pi(1 - \pi))/n}$, or in this case, $\sqrt{(0.51 * 0.49)/100} = 0.050$. (Note this uses the known population proportion π since we know the ultimate result here is $\pi = 0.51$). The probability of drawing a random sample in which the other candidate (Happy) comes out on top with 51 percent or more of the survey respondents favoring that candidate is the same as the probability that *49 percent or fewer support Grumpy*. Applying the CLT, the sample proportion will have an approximate

normal distribution with a large enough n . So:

$$\begin{aligned} P(\hat{\pi} \leq 0.49) &= P\left(\frac{\hat{\pi} - 0.51}{0.050} \leq \frac{0.49 - 0.51}{0.050}\right) \\ &= P(z \leq -0.4) \\ &= 0.345 \end{aligned}$$

In other words, there is a 34.5% chance that a random sample of 100 would show 49% or less support for Grumpy if the underlying population proportion was 51%. One could also solve this by defining $\pi = 0.49$ as the population proportion supporting Happy, and then finding $P(\hat{\pi} \geq 0.51)$. The result would be the same.

5. **(18 points)** The Chi-squared (χ^2) distribution is a common probability distribution in statistics defined by one parameter k (the degrees of freedom). If $x \sim \chi^2(k)$, then $E(x) = k$ and $Var(x) = 2k$. The skewness of the distribution is $\sqrt{8/k}$. For large values of k , the distribution is symmetric. For smaller values of k , it is positively skewed. For this problem, let $k = 10$. Create a Stata do file that does the following:

See Stata results attached below.

- (2 points)** Draw 200 random values from this Chi-squared distribution. Note Stata has a random number function for the Chi-squared distribution.
- (2 points)** Create a histogram for your sample data and find the sample mean $\bar{x}$ and standard deviation s . How do these compare to the (known) population mean and standard deviation? Describe the shape of your distribution: is it symmetric or skewed? What is the skewness statistic?
- (2 points)** Following one of the methods shown in class, conduct a simulation that repeatedly samples 10 observations from the Chi-squared distribution (with $k = 10$ as before), a total of 100 times. Your simulation should store the sample mean $\bar{x}$ on each iteration. When complete, use your data to answer the next questions.
- (4 points)** Create a histogram for these simulated sample means. What is the mean of these values? What is the standard deviation? How do these compare to what you would have predicted them to be, before you drew any samples? Explain.
- (2 points)** Based on your simulated sampling distribution, what is the probability of drawing a sample with a $\bar{x}$ greater than 11?
- (6 points)** Repeat parts (c)-(e) but increase the random sample size to 50. How

does this change your results?

6. (5 points) Consider this probability distribution for student absences from the last problem set:

# of Days	0	1	2	3	4	5	6	7	8
Probability	0.2	0.14	0.25	0.11	0.1	0.09	0.05	0.03	0.03

Now imagine you were to draw random samples of 50 students repeatedly from this population. What would the sampling distribution of $\bar{x}$ look like? What would be its mean? Its standard error? How do you know?

This population is definitely not a normal distribution. However, the CLT tells us that if the sample size is large enough, $\bar{x}$ will have an approximately normal sampling distribution with a mean of μ and a standard error of $\sigma/\sqrt{n}$. From the last problem set, $\mu = E(X) = 2.57$ and $\sigma = \sqrt{Var(X)} = \sqrt{4.57} = 2.14$ so that the standard error would be $2.14/\sqrt{50} = 0.302$.

7. (6 points—2 each) In Stata, use the code below to create a “population” distribution that reflects the probability distribution in part (6):

See Stata results attached below.

- What is the mean, median, variance, and standard deviation of this population?
- Use the `bootstrap` prefix to draw 1,000 repeated samples of size $n = 30$ from this population, each time retaining the mean, median, and standard deviation. (Hint: you will `bootstrap` the `summarize, detail` command). Save the results in a new dataset.
- Provide histograms for your resulting means, medians, and standard deviations. Across your 1,000 samples, what the *average* mean, median, and standard deviation? For this simulation, what is the *standard error* of the mean, median, and standard deviation? Explain in words what this quantity represents.

```

. // *****
. // LPO.8800 Problem Set 4 - Solutions to Questions 1, 5 and 7
. // Last updated: September 28, 2026
. // *****
.
.       cd "$pset"
C:\Users\corcorssp\Dropbox\_TEACHING\Statistics I - PhD\Problem sets\Problem set
> resources\Problem set 4

. // *****
. // Question 1
. // *****
.
. // In population what proportion fall between 80 and 90? 0.645
. display normal((90-83)/5) - normal((80-83)/5)
.64499022

. // Now generate own data using 1,000 random draws from this population
. set seed 1005

. set obs 1000
Number of observations (_N) was 0, now 1,000.

. gen midterm = rnormal(83, 5)

. histogram midterm, title(Draws from Normal(83,5) distribution) ///
> name(midterm, replace)
(bin=29, start=67.000526, width=1.0537246)

. graph export Q1midterm.pdf, as(pdf) replace
file Q1midterm.pdf saved as PDF format

. gen gradeb = midterm>=80 & midterm<=90

. summ gradeb

Variable | Obs      Mean      Std. dev.      Min      Max
-----+-----
gradeb  | 1,000    .629      .4833142      0        1

. // In my case, 62.9% of scores fall between 80 and 90 (inclusive). (Your
. // results will vary due to the random draws). This differs from the known
. // proportion in the population that fall between 80-90, found above (64.5%).
. // They differ because the generated data represent one, albeit large, random
. // sample from the population.
.
. // *****
. // Question 5
. // *****
.
. // *****
. // Part a-b
. // *****
.

```

```

.      set seed 5001

.      clear

.      set obs 200
Number of observations (_N) was 0, now 200.

.      gen x=rchi2(10)

.      histogram x, title(Draws from Chi-squared(10) distribution) name(chi2x
> , replace)
(bin=14, start=1.4292891, width=1.9391582)

.      graph export Q5chi2x.pdf, as(pdf) replace
file Q5chi2x.pdf saved as PDF format

.      sum x,detail

```

----- x -----				
	Percentiles	Smallest		
1%	2.226934	1.429289		
5%	3.870594	2.22456		
10%	5.071997	2.229308	Obs	200
25%	7.195452	2.817673	Sum of wgt.	200
50%	9.874336		Mean	10.55522
		Largest	Std. dev.	4.753164
75%	13.54217	24.09533		
90%	16.45877	24.25708	Variance	22.59257
95%	19.22234	24.37455	Skewness	.8102286
99%	24.31581	28.5775	Kurtosis	3.811873

```

.      // In my case, the sample mean and variance are 10.555 and 22.593, respectively
> y.
.      // (Your results will vary due to the random draws). The known population mean
>
.      // of this distribution is  $E(x)=k = 10$ , which is close to my sample mean.
.      // The known population variance is  $Var(x)=2k = 20$ , which is again close to
.      // my sample variance. The distribution is right-skewed.
.      // See the descriptive statistics and histogram.

.      // *****
.      // Part c
.      // *****
.      // Simulate 100 samples of n=10 from Chi-squared(10) distribution

.      capture program drop chi2

.      program chi2, rclass
1.          drop _all
2.          set obs 10
3.          gen x=rchi2(10)
4.          summ x
5.          return scalar mean=r(mean)
6.      end

.      simulate mean=r(mean) , reps(100) nodots: chi2

Command: chi2
mean: r(mean)

```

```

. // *****
. // Part d
. // *****
. // View sampling distribution (100 samples)
.
.       histogram mean, title(Sampling distribution of x-bar n=10) ///
>       name(xbar10,replace)
(bin=10, start=7.1717663, width=.65956135)

.       graph export Q5xbar10.pdf, as(pdf) replace
file Q5xbar10.pdf saved as PDF format

.       summ mean, det

               r (mean)
-----
Percentiles   Smallest
1%      7.215956   7.171766
5%      7.855759   7.260146
10%     8.320474   7.688992   Obs           100
25%     9.157813   7.764164   Sum of wgt.    100

50%     10.1359
               Largest      Mean           10.21342
75%     11.1312   12.91886   Std. dev.    1.501115
90%     12.45014   13.24501   Variance     2.253347
95%     12.87675   13.43837   Skewness     .2824596
99%     13.60288   13.76738   Kurtosis     2.48551

. // In my case the mean of the sample means is 10.21 and the standard deviation
>
. // is 1.50. The Central Limit Theorem tells us that the mean of sampling
. // distribution of x-bar will be mu, or 10 in this case. The true (population)
. // standard deviation of the sampling distribution of x-bar (the standard
. // error) will be sigma/sqrt(n), or sqrt(20)/sqrt(10)= 1.41 in this case.
.
. // *****
. // Part e
. // *****
. // What fraction of sample means exceed 11?
.
.       count if mean>11
.29

.       display `r(N)'/_N

. // We can look directly at the resulting sampling distribution to find the
. // proportion of times x-bar is greater than 11. In my case this is 29%.
.
.
. // *****
. // Part f
. // *****
. // Repeat parts c-e with sample size of 50
.
.       capture program drop chi2

```

```

.      program chi2, rclass
1.          drop _all
2.          set obs 50
3.          gen x=rchi2(10)
4.          summ x
5.          return scalar mean=r(mean)
6.          end

.      simulate mean=r(mean) , reps(100) nodots: chi2

Command: chi2
       mean: r(mean)

.      histogram mean, title(Sampling distribution of x-bar n=50) name(xbar50
> ,replace)
(bin=10, start=8.5110826, width=.27406311)

.      graph export Q5xbar50.pdf, as(pdf) replace
file Q5xbar50.pdf saved as PDF format

.      summ mean,det

               r(mean)
-----
Percentiles      Smallest
1%      8.543072      8.511083
5%      9.061373      8.575061
10%     9.227685      8.753469      Obs      100
25%     9.653508      9.002769      Sum of wgt. 100

50%     10.15116
75%     10.54641      Largest      Mean      10.08912
90%     10.92034      11.12409      Std. dev.  .6313759
95%     11.09847      11.14734      Variance   .3986355
99%     11.22367      11.19563      Skewness   -.2275299
          11.25171      Kurtosis    2.444882

.      count if mean>11
      8

.      display `r(N)'/_N
      .08

.      // The result of a larger sample size is a sampling distribution with a smalle
> r
.      // standard error (now 0.63 instead of 1.5). The theoretical standard error
.      // for a sample size of 50 is sqrt(20)/sqrt(50) = 0.632. In this new setting
.      // the probability of drawing an x-bar>0.60 has fallen to 0.08 (i.e. such an
.      // extreme value is less likely).
.
.      // *****
.      // Question 7
.      // *****
.      // Create a population distribution that looks like the probability distributi
> on

```

```

. // in #6 (assuming 100 in the population for simplicity).
.
.      clear all
.
.      set obs 20
Number of observations (_N) was 0, now 20.
.
.      gen abs = 0
.
.      insobs 14 /* insobs inserts additional observations */
(14 observations added)
.
.      replace abs = 1 if abs==.
(14 real changes made)
.
.      insobs 25
(25 observations added)
.
.      replace abs = 2 if abs==.
(25 real changes made)
.
.      insobs 11
(11 observations added)
.
.      replace abs = 3 if abs==.
(11 real changes made)
.
.      insobs 10
(10 observations added)
.
.      replace abs = 4 if abs==.
(10 real changes made)
.
.      insobs 9
(9 observations added)
.
.      replace abs = 5 if abs==.
(9 real changes made)
.
.      insobs 5
(5 observations added)
.
.      replace abs = 6 if abs==.
(5 real changes made)
.
.      insobs 3
(3 observations added)
.
.      replace abs = 7 if abs==.
(3 real changes made)
.
.      insobs 3
(3 observations added)
.
.      replace abs = 8 if abs==.
(3 real changes made)
.
.      tabulate abs

```

abs	Freq.	Percent	Cum.
0	20	20.00	20.00
1	14	14.00	34.00
2	25	25.00	59.00
3	11	11.00	70.00
4	10	10.00	80.00
5	9	9.00	89.00
6	5	5.00	94.00
7	3	3.00	97.00
8	3	3.00	100.00
Total	100	100.00	

```

.
. // *****
. // Part a
. // *****
.
.       summ abs, detail

-----+-----
Percentiles      Smallest
1%              0          0
5%              0          0
10%             0          0      Obs          100
25%             1          0      Sum of wgt.    100

50%             2
75%             4          7      Mean          2.57
90%             6          8      Std. dev.     2.147373
95%             7          8      Variance     4.611212
99%             8          8      Skewness     .7216121
                        Largest      Kurtosis     2.769535

. // The mean is 2.57, the median is 2, the variance is 4.61 and the standard
. // deviation is 2.15. If these data represent the population, then these are
. // the population parameters.
. // *****
. // Part b
. // *****
.
.       bootstrap r(mean) r(p50) r(sd), size(30) reps(1000) ///
>       saving(absresults,replace): summ abs,detail
(running summarize on estimation sample)

warning: summarize does not set e(sample), so no observations will be excluded
        from the resampling because of missing values or other reasons. To
        exclude observations, press Break, save the data, drop any
        observations that are to be excluded, and rerun bootstrap.

Bootstrap replications (1,000): .....10.....20.....30.....40....
> .....50.....60.....70.....80.....90.....100.....110....
> .....120.....130.....140.....150.....160.....170.....
> 180.....190.....200.....210.....220.....230.....240....
> .....250.....260.....270.....280.....290.....300.....
> 310.....320.....330.....340.....350.....360.....370....
> .....380.....390.....400.....410.....420.....430.....
> 440.....450.....460.....470.....480.....490.....500....
> .....510.....520.....530.....540.....550.....560.....
> 570.....580.....590.....600.....610.....620.....630....
> .....640.....650.....660.....670.....680.....690.....
> 700.....710.....720.....730.....740.....750.....760....
> .....770.....780.....790.....800.....810.....820.....
> 830.....840.....850.....860.....870.....880.....890....
> .....900.....910.....920.....930.....940.....950.....
> 960.....970.....980.....990.....1,000 done

Bootstrap results                                Number of obs =   100
                                                Replications = 1,000

Command: summarize abs, detail
       _bs_1: r(mean)
       _bs_2: r(p50)
       _bs_3: r(sd)

```

	Observed	Bootstrap			Normal-based	
	coefficient	std. err.	z	P> z	[95% conf. interval]	
_bs_1	2.57	.3907242	6.58	0.000	1.804195	3.335805
_bs_2	2	.4159904	4.81	0.000	1.184674	2.815326
_bs_3	2.147373	.2642287	8.13	0.000	1.629495	2.665252

```

.
.       use absresults, clear
(bootstrap: summarize)

.       rename _bs_1 xbar

.       rename _bs_2 xmedian

.       rename _bs_3 xstddev

.
.       summ xbar xmedian xstddev

Variable |           Obs           Mean      Std. dev.        Min        Max
-----+-----
      xbar |           1,000      2.565367      .3907242      1.433333      3.866667
    xmedian |           1,000           2.125      .4159904           1           4
    xstddev |           1,000      2.120229      .2642287      1.407696      2.932968

.
.       histogram xbar, name(xbarabs, replace) nodraw
(bin=29, start=1.4333333, width=.08390804)

.       histogram xmedian, name(xmedabs, replace) nodraw
(bin=29, start=1, width=.10344828)

.       histogram xstddev, name(xstddev, replace) nodraw
(bin=29, start=1.4076964, width=.05259556)

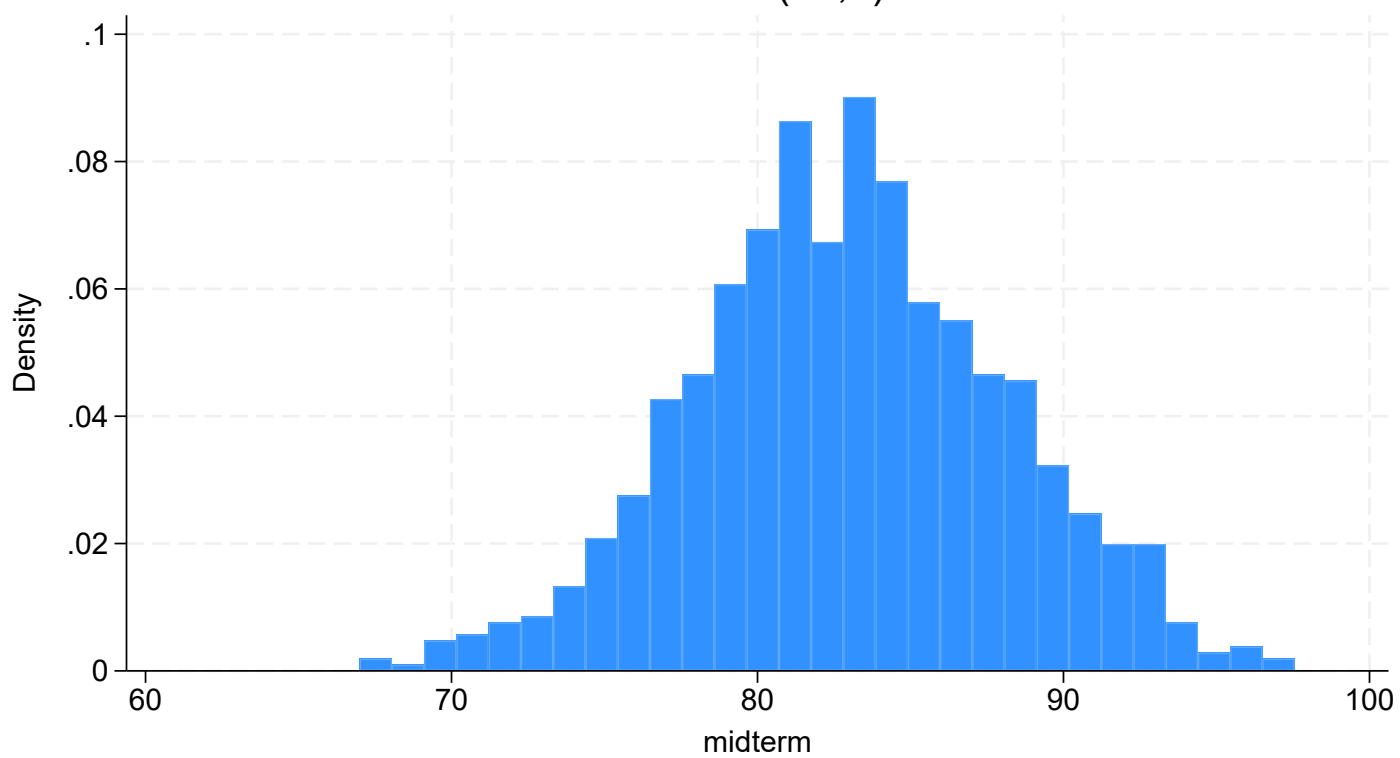
.       graph combine xbarabs xmedabs xstddev, col(1) xsize(3) ysize(6)

.       graph export Q7.pdf, as(pdf) replace
file Q7.pdf saved as PDF format

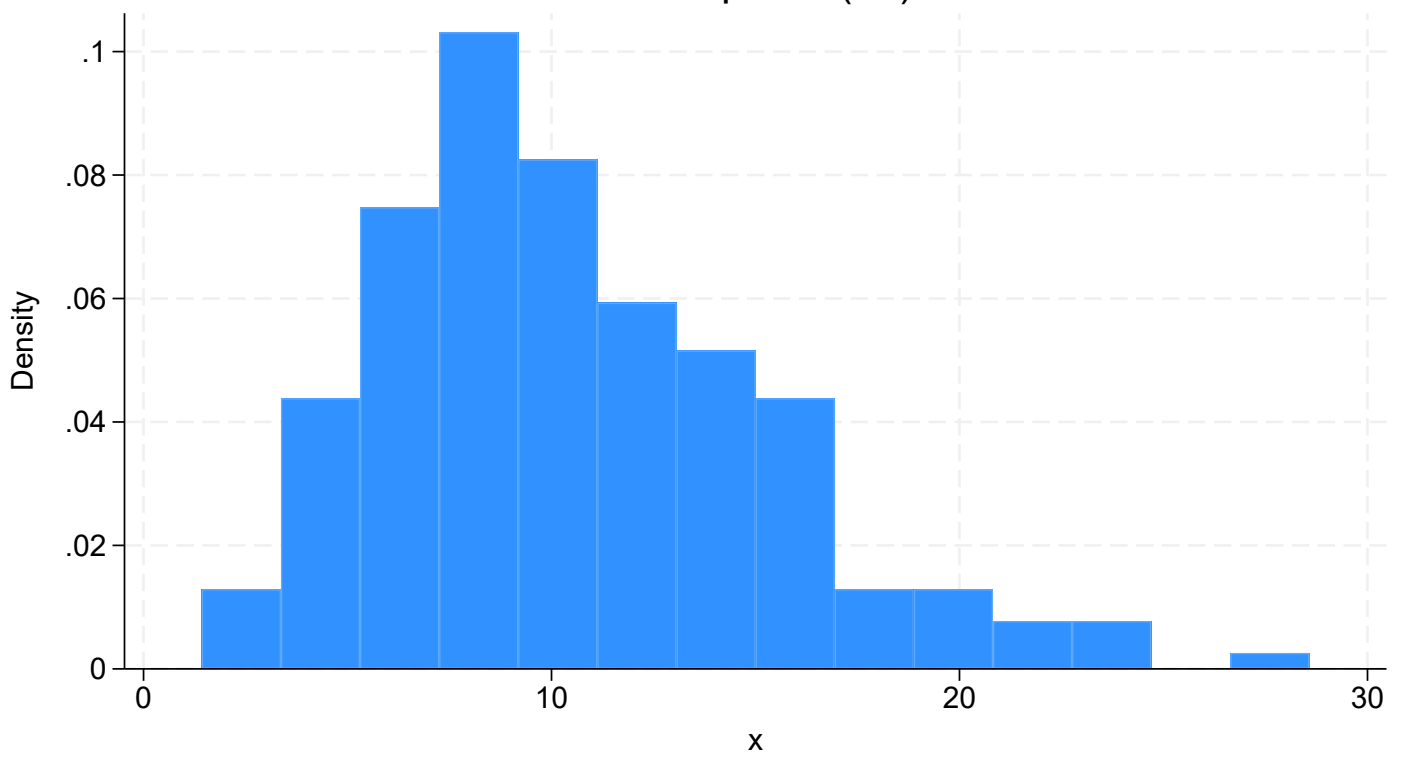
.
. // The histograms (sampling distributions) are shown. The average value
. // for the mean (xbar) was 2.57 with a standard deviation (std error) of 0.391
> .
. // The average value for the median was 2.13 with a standard deviation (std
. // error) of 0.416. The average value for the std deviation was 2.12 with a
. // std deviation (standard error) of 0.264. These are good examples of three
. // statistics that vary from sample to sample. These simulated sampling
. // distributions give us a sense of their average value across samples and how
> .
. // much they vary from the average from sample to sample. (The std error
. // captures that).
.
. // Close log and convert to PDF
.       capture log close

```

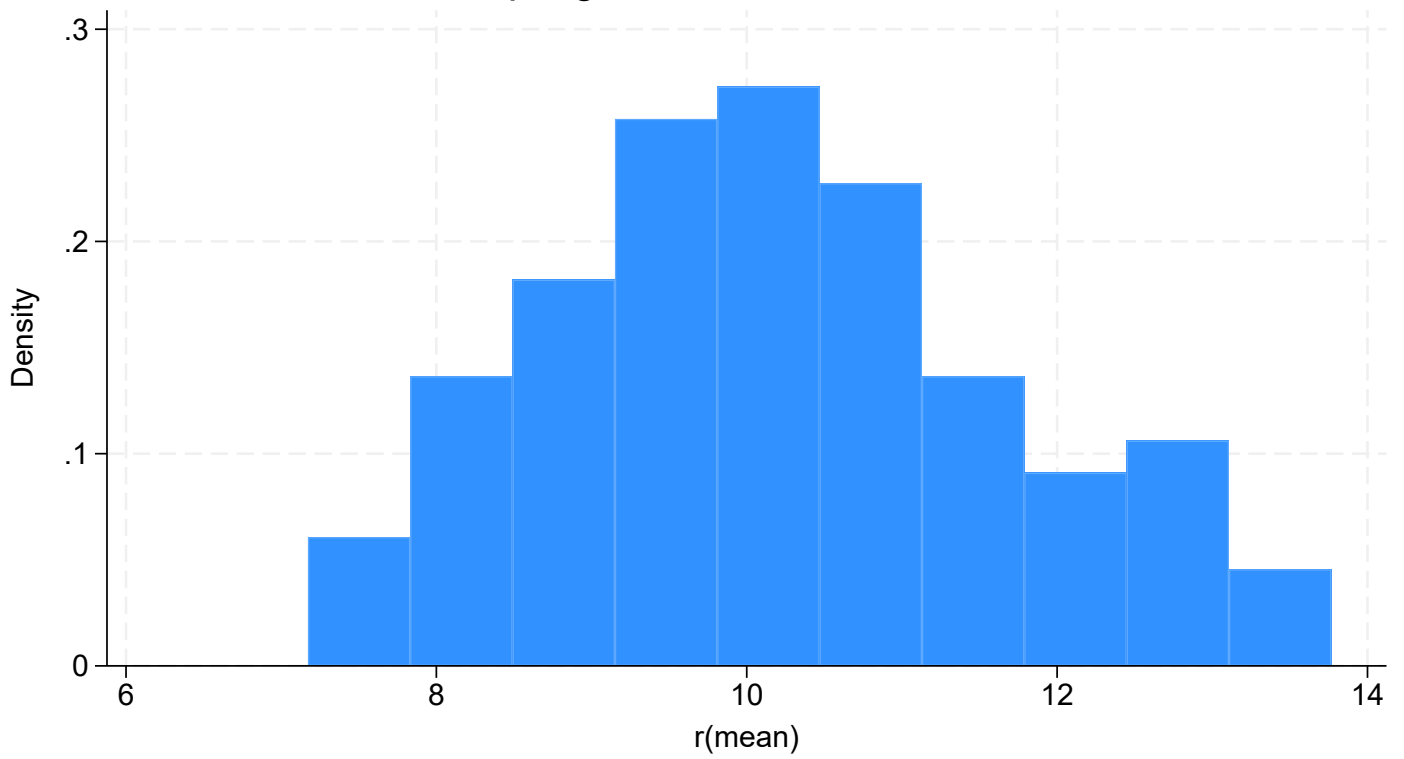
Draws from Normal(83,5) distribution



Draws from Chi-squared(10) distribution



Sampling distribution of $\bar{x}$ n=10



Sampling distribution of $\bar{x}$ n=50

